

Irfan Tariq, Ph.D.

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🌐 <https://scholar.google.com/citations?user=QCzXbwcAAAAJ&hl=en>

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Education

- Sep 2018 – June 2023 **Ph.D., Southeast University, Nanjing, China.**
Major: *Information and Communication Engineering.*
Thesis title: *Researches in the detection and classification of Pulsar signals*
Courses study: *Advance Signal Analysis and processing , Detection and Estimation, VLSI Design , IC for Optic-Fiber Communication .*
- Sep 2014 – March 2017 **Ms, Nanjing University of Science and Technology, Nanjing, China.**
Courses study: *Digital signal processing , Digital Communication, Software Radio Technology , Advance Signal Processing , Digital image processing, Wireless Sensor Network.*
Major: *Signal and Information processing.*
Thesis title: *Wideband Sensing for Sparse Spectrum.*
- Feb 2010 – March 2014 **Bs, COMSATS Institute of Information Technology, Wah, Pakistan.**
Major: *Bachelor of Science in Electrical Telecommunication Engineering .*
Courses study: *Digital Logic Design, Data Communication and Computer Networking, Digital Communication System, Wireless Communication System , Digital image processing, Wireless Sensor Network.*

Employment History

- October 2023 – Present **Staff Engineer, Synopsys Semiconductor Technology (Shanghai) Co., Ltd**
Key Responsibilities and Contributions:
- Performed Clock Tree Synthesis (CTS) using industry-leading EDA tools such as ICCII and Fusion Compiler for advanced digital SoC designs.
 - Optimized clock distribution networks to achieve balanced latency, reduced insertion delay, improved timing quality, and lower power consumption.
 - Analyzed clock tree topology, buffering strategies, and routing structures to ensure robust clock propagation across complex designs.
 - Conducted detailed timing analysis including setup/hold closure, clock skew optimization, and Multi-Corner Multi-Mode (MCMM) verification.

Employment History (continued)

- Drove CTS-based Power, Performance, and Area (PPA) optimization to improve timing convergence, clock power efficiency, routing congestion, and overall design robustness.
- Debugged complex CTS issues such as excessive skew, transition violations, IR-drop hotspots, and noise coupling through in-depth root-cause analysis.
- Implemented and refined clock gating strategies to reduce dynamic power while maintaining timing margins and functional correctness.
- Analyzed CTS and post-CTS timing reports, RC extraction data, and violation patterns to identify root causes and implement effective solutions.
- Validated the end-to-end RTL-to-GDSII flow, from pre-CTS preparation through post-route timing closure, ensuring signoff readiness.
- Designed comprehensive validation and test plans based on product specifications, customer requirements, and feature documentation.
- Reproduced and resolved customer-reported issues, delivering timely technical solutions and recommendations.
- Performed QoR benchmarking and analysis on industry-standard designs targeting advanced technology nodes (e.g., 3nm, 5nm).
- Developed and maintained automated regression frameworks, validation suites, and CI pipelines to ensure tool stability and quality.
- Created and optimized automation scripts (Python, Tcl, Shell) for test execution, report generation, data processing, and large-scale PPA analysis.
- Collaborated closely with R&D, Applications Engineering (AE), and Product Management (PM) teams to drive defect resolution, feature enhancements, and tool optimization.

Research Publications

Journal Articles

- 1 TARIQ, I., DONG, N.-f., WANG, J.-x., & ABBAS, S. R. (n.d.). Fast computing dft method for sparse signals based on downsampling.
- 2 Khan, A., Tariq, I., Khan, H., Khan, S. U., He, N., Zhiyang, L., Raza, F. et al. (2022). Lung cancer nodules detection via an adaptive boosting algorithm based on self-normalized multiview convolutional neural network. *Journal of Oncology*, 2022.
- 3 Li, S., Meng, Q., Tariq, I., & Chen, X. (2022). A 78-mhz bw continuous-time sigma-delta adc with programmable vco quantizer. *CMC-COMPUTERS MATERIALS & CONTINUA*, 72(3), 6079–6090.

- 4 Liu, W., Meng, Q., Wang, C., Zhou, C., Yao, S., & Tariq, I. (2022). A real-time, pipelined incoherent dedispersion method and implementation in fpga. *Publications of the Astronomical Society of the Pacific*, 134(1031), 015008.
- 5 Tariq, I., Qiao, M., Wei, L., Yao, S., Zhou, C., Ali, Z., ... Spanakis-Misirliis, A. (2022). Classification of pulsar signals using ensemble gradient boosting algorithms based on asymmetric under-sampling method. *Journal of Instrumentation*, 17(03), Po3020.
- 6 Tariq, I., Meng, Q., Yao, S., Liu, W., Zhou, C., Ahmed, A., & Spanakis-Misirliis, A. (2022). Adaboost-dsnn: An adaptive boosting algorithm based on deep self normalized neural network for pulsar identification. *Monthly Notices of the Royal Astronomical Society*, 511(1), 683–690.
- 7 Yao, S., Meng, Q., Chen, C., & Tariq, I. (2022). A dptf algorithm for the time-delay estimation in the reflected environment. *Digital Signal Processing*, 127, 103534.
- 8 Liu, W., Meng, Q., Wang, C., Zhou, C., Yao, S., & Tariq, I. (2021). An efficient channelization architecture and its implementation for radio astronomy. *Journal of Instrumentation*, 16(08), Po8047.
- 9 Yao, S., Meng, Q., Chen, C., Tariq, I., Zhou, C., & Liu, W. (2021). High-precision time delay estimation of narrowband radio signal by phat-lstm. *Measurement Science and Technology*, 32(7), 075001.
- 10 Azeem, S. W., Chen, W., Tariq, I., Ye, H., & Kaija, D. (2020). A hybrid resonant zvzcs three-level converter suitable for photovoltaic power dc distribution system. *IEEE Access*, 8, 114981–114990.
- 11 Tariq, I., & Qiao, M. (2019). A fast dft method for generally k sparse signals recovery. *SN Applied Sciences*, 1, 1–6.
- 12 Iqbal, M., Zhang, K., Iqbal, S., & Tariq, I. (2018). A fast and reliable dijkstra algorithm for online shortest path. *Int. J. Comput. Sci. Eng*, 5(12), 24–27.

Conference Proceedings

- 1 Tariq, I., Qiao, M., Yao, S., Ullah, K., Khan, S. U., & Wei, L. (2022). Cost sensitive self-normalized deep convolutional neural network for pulsars selection. In *2022 4th international conference on computer communication and the internet (iccci)* (pp. 116–121). IEEE.
- 2 Khan, A., He, N., Tariq, I., & Li, Z. (2021). Stacking ensemble method for early and advanced stage lung adenocarcinoma classification based on mirna expression. In *2021 10th international conference on bioinformatics and biomedical science* (pp. 76–81).
- 3 Khan, A., Baohua, W., Ahmed, S. B., & Tariq, I. (2020). A current transformer saturation time difference method for detecting out of zone fault using mathematical morphology. In *2020 ieee 1st china international youth conference on electrical engineering (ciyce)* (pp. 1–5). IEEE.
- 4 Sharif, U., Sun, B., Tariq, I., Ibrahim, D. S., Adewale, O. O., & Zafar, A. (2020). Static and modal analysis of sandwich beam structure with magnetorheological honeycomb core. In *2020 6th international conference on mechanical engineering and automation science (icmeas)* (pp. 19–23). IEEE.
- 5 Shah, S. S., Sun, W., Tariq, I., & Xu, S. (2019). Analog compensator design for half bridge llc resonant converter. In *2019 ieee 4th international conference on signal and image processing (icsip)* (pp. 313–317). IEEE.
- 6 Abbas, Z., Li, J., Yadav, N., & Tariq, I. (2018). Computational task offloading in mobile edge computing using learning automata. In *2018 ieee/cic international conference on communications in china (iccc)* (pp. 57–61). IEEE.
- 7 Khan, A., Wang, B.-h., Tariq, I., & Abbas, S. R. (2017). A comparative study of morphological gradient and its improved form for ct saturation detection. In *Wcne* (pp. 364–368).

Skills

- Languages
 - Strong reading, writing and speaking competencies for English.
 - Basic Chinese
 - Mother tongue: Pashto and Urdu.
- Coding
 - Matlab, Python, TensorFlow, PyTorch, TCL, Verilog, Linux.
 - Microsoft Office, PowerPoint.
- Misc.
 - Academic research, $\text{\LaTeX}$ typesetting and publishing.

Certification

- May 2022-June-2022
 - Convolutional Neural Network**
Awarded by deeplearning.ai

References

Available on Request